Reneta Trifonova

MSc Embedded Systems | Software Test Engineer

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SUMMARY

MSc Embedded Systems student at Eindhoven University of Technology with experience in software engineering, CI/CD automation, embedded systems and education. Experienced in solving engineering problems independently and collaborating across technical and customer-facing teams. Interested in applying technology to complex industrial and energy challenges.

EXPERIENCE

Software Test Engineer (Part-Time)

Novenda 11/2024 - Present Eindhoven, NL

- Designed and independently implemented multi-stage CI/CD pipelines for company software applications, automating builds, Google Test/Python testing and code analysis in secure virtual environments
- Strengthened software security by integrating automated vulnerability scanning and SBOM generation, analyzing 30+ software components and dependencies
- Identified bugs to the development team through executing acceptance testing against functional requirements

Product and Implementation Engineer (Part-Time)

Cofano Software Solutions 07/2022 - 05/2024 Enschede, NL

- Developed logistics software supporting container-terminal operations across the Netherlands, Germany, Belgium and the UK
- Bridged communication between terminal employees and developers, analysing customer issues and translating them into reproducible bug reports and technical requirements
- Automated container-booking imports across CSV, Excel, JSON and XML formats, eliminating repetitive manual data entry for terminal employees

Teaching Assistant (Part-Time)

University of Twente 09/2022 - 08/2024 Enschede, NL

- Developed and graded cryptography exercises on Arduino UNO for 90+ students in an Embedded Systems course
- Mentored students and helped lecturers with grading and teaching programming concepts in Java, C, Linux, SQL

PROJECTS

Multimodal Split Learning under Missing Modalities

11/2025 - Present

Individual MSc Thesis (Machine Learning, Python, PyTorch)

- Investigated the lack of robustness of multimodal transformer models under missing modalities in split-learning setup.
- Independently designed a Transformer framework to evaluate alternative approaches such as fusion and imputation techniques to improve accuracy.

Securing Cloud Infrastructure

08/2026 - 10/2026

Personal Project (Linux, Azure, Cloudflare, Docker, Bash, SSH, Prometheus, Grafana)

- Deployed and secured Azure-based Linux infrastructure with network segmentation, Cloudflare Zero Trust authentication, and firewall policies.
- Configured automated monitoring, secure remote access, and fault detection.

DIS Attack and Defense

02/2026 - 04/2026

Team Project (C, Linux, Contiki-NG, Wireshark, IPv6, RPL, IEEE 802.15.4, nRF52840, CC2650)

- Conducted DIS flooding and IPv6 attacks on a physical IoT network using the RPL routing protocol. Protected the network using rate-limiting and MAC-address-based filtering in Contiki-NG.
- Used Wireshark to validate the mitigation's effectiveness and prove the reduction in malicious traffic.

Lightweight Security Protocols for IoT networks

04/2024 - 07/2024

Individual BSc Thesis (ESP32, Raspberry Pi, Zephyr RTOS, CoAP, OSCORE, EDHOC, DTLS)

- Conducted research on lightweight IoT cybersecurity, implementing and evaluating OSCORE and EDHOC protocols on ESP32 and Raspberry Pi devices.
- Benchmarked authentication methods, hardware acceleration, memory usage, latency, and power consumption of the protocols.

EDUCATION

MSc Embedded Systems

Eindhoven University of Technology

09/2024 - Present Eindhoven, NL GPA 8.0 / 10

BSc Technical Computer Science and Engineering

University of Twente

09/2021 - 07/2024 Enschede, NL GPA 8.2 / 10

- Cum Laude Distinction, Best Paper Award

SKILLS

Cybersecurity & Network Security

Cloudflare Tunnels, Secure Remote Access, Access Control, Traffic Filtering, Vulnerability Scanning, SBOM, Wireshark, Packet Analysis

Programming

Python, Java, C, C++, C#, SQL

DevOps & Testing CI/CD

Azure DevOps, Azure Pipelines, Jenkins, Google Test, Docker, Prometheus, Grafana

Machine Learning

Python, PyTorch, NumPy, Jupyter, Transformers

Web Development

React, Vue, GraphQL, JavaScript, SQL

Embedded

Raspberry Pi, ESP32, Arduino, Soldering, Real-Time Programming, FPGA, Contiki-NG

LANGUAGES

English Proficient

Bulgarian Native

Dutch Advanced

INTERESTS

CTF

Competing in Capture the Flag Cybersecurity Challenges

Collecting Magazines

Learning about new topics through Natural Geographic Magazines

Learning new Languages

Learning to communicate in Dutch, German and French

Traveling

Discovering new destinations